Stress Detection using ECG-EEG

by Aarthi S

Submission date: 15-May-2026 09:22AM (UTC+0530)
Submission ID: 2961641755
File name: ECG-EEG_miniproject_paper.docx (266.87K)
Word count: 3141
Character count: 25160

An Explainable Multimodal Transformer Framework for EEG–ECG-Based Stress Detection with RAG-Enabled Personalized Intervention

Dr.AarathiS
Associate Professor
Department of CSE (Data Science)
DayanandaSagarCollege of Engineering
Bengaluru, India
aarthi-csds@dayanandasagar.edu

Ashlesh
DepartmentofCSE(DataScience)
DayanandaSagarCollege of Engineering
Bengaluru, India
ashlesh.kanchan@gmail.com

SambhramSNaregal
DepartmentofCSE(DataScience)
DayanandaSagarCollege of Engineering
Bengaluru, India
sambhramnaregal22@gmail.com

TusharDKudachimath
DepartmentofCSE(DataScience)
DayanandaSagarCollege of Engineering
Bengaluru, India
1ds23cd058@dsce.edu.in

VishalKrishnaBhatK
DepartmentofCSE(DataScience)
DayanandaSagarCollege of Engineering
Bengaluru, India
vkb39004@gmail.com

***Abstract*—Stress is a prevalent issue worldwide which impacts one's psychological and physical health. Recent developments have been made in the detection of stress through wearables which measure physiological signals such as Electroencephalography (EEG), Electrocardiogram (ECG), Electrodermal Activity (EDA), Galvanic Skin Response (GSR), and Photoplethysmography (PPG). Machine learning models utilized for prediction include Logistic Regression, KNN, Random Forest, Support Vector Machines (SVM), XGBoost, and Neural Networks. In order to provide an explanation and make the predictions more understandable, Explainable Artificial Intelligence (XAI) methodslikeSHAP,LIME,andattentionmechanismsareutilized.**

***Index Terms*—Recognition, Wearable Technology, Machine Learning, Explainable Artificial Intelligence, Physiological Data, Electroencephalogram,Electrocardiogram,ElectrodermalActivity, Heart Rate Variability, Biosignal Processing, Mental WellbeingSurveillance,HealthCareDataAnalysis,DeepLearning**

I. Introduction

The importance of detecting stress and monitoring mental health conditions has become an essential area of researchdue to the fast-growing number of psychological disordersand illnesses which are associated with people's lifestyles. In today'sera,individualsfaceconstantchallengesineducational institutions, workplaces, and the environment, causing stress that adversely impacts their physical and psychological state. Stressisincreasinglyacknowledgedasasignificantfactor in developing such ailments as high blood pressure, obesity, diabetes,heartdiseases,andmetabolicsyndromes[5].Consideringthatcurrentclinicaldiagnosticsrelyprimarilyonselfreports and laboratory analyses, there is an urgent need for an automated monitoring system for detecting stress.

Technological developments in the domain of wearable sensing have made it possible to monitor in a continuous manner the biosignals related to stress response mechanisms. SignalslikeEEG,ECG,EDA,GSR,andPPGrepresent someofthemostdirectmeasuresforassessingtheactivi- ties of the autonomic nervous system. In research involving wearable EEG, GSR, and PPG signals, accurate classification was achieved for chronic mental stress conditions in which neurophysiological features such as theta EEG were found to be an important contribution [1]. Multimodal data sets like CLAREinvolvedECG,EEG,EDA,andeyetrackingmeasures and revealed that stress classification improved by employing several physiological measures [2].

The application of machine learning algorithms has improved stress recognition systems through automated learning of features from physiological data. In the study related to early cognitive load anticipation, ECG and EEG signals were used together with features like relative power, coherence, phase difference, and asymmetry of amplitude, resulting in successful classification via Support Vector Machines and dimensionality reduction [3]. Furthermore, investigations about brain-heart interaction during mental and physical stresses showed that physiological complexity varies according to stress stimuli, and thus it is essential to examine the relationship between physiological processes dynamically rather than statically [4].

Due to the increased usage of wearable gadgets like smartwatchesandsmartrings,ithasbecomepossibletomonitor one’shealthstatusinreal-timebeyondlabsettings.Biosignals, whichincludeheartrate,sweatingresponse,andelectrodermal activity, can be measured automatically by such wearables to help people determine their stress levels continuously without the need for any special device [5]. Various machine learning algorithms have been found to produce impressive results whileprocessingbiosignaldata,someofwhichincludelogistic regression,K-nearestneighbour,randomforest,supportvector machines, XGBoost, and neural networks. However, most of these models remain black boxes with little interpretability, making them unsuitable for clinical settings. Thus, scientists have proposed XAI methods such as SHAP, LIME, and attention mechanisms [6].

Further advancements incorporate explainability principles into the sphere of cardiovascular and medicine diagnostics.Forexample,deeplearningtechniquesbasedonconvo lutional neuralnetworkshavebeenusedtodiagnosecardiovascu- lar diseases through electrocardiogram measurements basedon waveform portions that impact their predictive decision-making capabilities [7]. Additionally, with regard to heart rate variability measurements, methods for explainable artificial intelligence help train machine learning techniques that can detect the time-related dynamics of cardiovascular diseases and the autonomic nervous system. Thus, with the help of these techniques, it becomes possible to extract essential physiological characteristics and thus obtain accurate clini-cal information. [8] In summary, the combination of recent wearable sensor advancements, advanced machine learning, andtheusageofXAIapproachesconstituteanimportant step in developing intelligent stress monitoring systems. The developmentandimplementationofthediscussedtechnologies and approaches allow creating new healthcare solutions that could reliably improve stress level evaluation, diagnosis, and medical treatment.

II. LITERATUREREVIEW

Majid et al. [1] suggested a technique to classify chronic mental stress types using wearable EEG, GSR, and PPG data which was taken from 40 people. With a wrapper based LFF method,theyfoundthatthetaEEGwasthemostimportantfea- ture for detecting stress. Using leave-one-out cross-validation, they chose MLP over SVM and Naive Bayes to get higher accuracy: 95% (F-measure 0.95) for two stress types and 77.5% (F-measure 0.75) for three.

The CLARE dataset [2] consisted of multimodal data col- lectedfrom24participantsduringtheexecutionoftheMATB-IItask.Thefollowingmodalitiesareincludedinthisdatabase: ECG, EDA, EEG, and eye-tracking signals, all collected at onceforeachparticipant.Participantsprovideratingsoftheir subjectivefeelingsofcognitiveloadonaLikertscaleevery10 seconds, which was used for labelling the data. The authors employedbothmachinelearninganddeeplearningalgorithms fortheclassification.ECG,EDA,andEEGachievedthe best performance when combined, with accuracy of 85.6% using10-foldcrossvalidation.Forleave-one-subject-outcross validation, accuracy was 72.7

Early prediction of cognitive load and psychological stress in [3] has been done based on analysing the ECG and EEG signalsviatheMISTtechniquebyexperimentingon22males. They have extracted features like relative power, coherence, phase lag, and amplitude asymmetry from 21 ECG and 40 EEG channels. In selecting the features, they utilized t-test pairedandPCAtoreducedimensions.ByapplyingSVMwith the radial basis function, their success rate was at 79.54%, sensitivity at 81%, and specificity at 78%. Their significant features were EEG relative power at the frontoparietal and EEG functional connectivity among electrodes. Possible limi- tationsincludetheall-malepopulationanda128-channelEEG that is highly complicated.

Catramboneetal.[4]researchedontherelationshipbetween BHI under mental and physical stress using mental arithmetic test and cold pressor test as the two stimuli. They measured EEGandECGdatainhealthyparticipants.Basedonthe SDG framework and Fuzzy Entropy, the paper showed that BHI complexity response to stress depends on the direction and stimulus type of stress. Thus, the mental stress promoted bottomupcomplexitywhilephysicalstresssawadecline in top down and an increase in bottom-up complexity . The sympathovagalLFpowerregulatesthiseffectofstressonBHI complexity.

Stress represents a very significant health issue which con- nects directly to obesity and hypertension and diabetes and various other metabolic disorders. The world in the current scenario experiences widespread stress which negatively im- pacts both human health and the economic system [5]. The humanbodyrespondstostressbyactivatingitsstressresponse system which results in an increased heart rate and sweating that medical professionals can detect through electrodermal activity (EDA) measurements. Modern wearable devices can be used by people which includes smartwatches and rings to automaticallymeasuretheirheartrateandEDAandotherbody signalsthathelpthemtracktheirstresslevelswithoutrequiring laboratory equipment.

Nnadi et al. [6], in his paper emphasizes Machine learning techniques which use biosignals like heart rate and electroder- malactivity(GSR)askeystressindicators.Supervisedmodels which include Logistic Regression, KNN, Random Forest, SVM, XGBoost and Neural Networks achieve precise perfor- mance in stress detection. These models use random sampling along with cross-validation methods for their training process. Theblackboxbehaviorofmultiplemodelsnecessitatestheuse of XAI techniques which includes SHAP, LIME and attention based systems to improve the prediction understanding in practical situations.

In yet some other systems Explainable AI methods areused to detect heart disorders through automated analysis of electrocardiogram(ECG)signals.Theresearchteamuseddeep learning models which included convolutional neural network architectures in order to extract spatio-temporal patterns from ECG data resulting in high diagnostic accuracy across differ- ent datasets. The research team used different explainability methodstoidentifyimportantwaveformsegmentsthataffected predictions and helped them solve the problems associated usually with the black-box model systems. The system also delivers precise predictions while providing interpretable decision processes which results in trustworthy clinical support and thus encourages medical facilities to adopt artificial intelligence based technologies [7].

The researchers in their paper [8], used an Explainable Artificial Intelligence (XAI) framework to analyze Heart Rate Variability (HRV) obtained from Electrocardiogram (ECG) signals for enhanced cardiovascular assessment. The researchers used machine learning models to study complex temporal patterns and found that in HRV data which helped them understand how cardiac health links to autonomic nervous system function. The study applied various explainability techniques to highlight the most influential physiological features affecting model predictions, and thereby improving model transparency and trustworthiness. The research developedvarioussystemsforcontinuouscardiachealthmonitoring which indeed provided accurate and reliable results through HRV analysis and interpretable AI methods that supported medical decision-making in clinical environments.

Lyberatosetal.[9]arguethattheincreasingdemandfor transparency and trust in medical artificial intelligence(AI) systems has rendered explainable artificial intelligence (XAI)essentialforelectroencephalogram(EEG)analysis.The EEG signal is characterized by its noise-prone nature, nonstationarity, and complexity, which makes it hard to interpretit using conventional analysis techniques. While the use of deepneuralnetworksforanalyzingsuchdataallowsobtaining valuable insights, their application is restricted due to the widespread practice of representing them as "black boxes." The study argues that interpretability is a critical factor for implementing AI applications within the medical industry, especially those involving emotion recognition or epilepsy detection tasks. The paper provides an in-depth overview of existing approaches to XAI used in conjunction with EEG studies,groupingthemaccordingtomodeldependency,explanation technique, and extent. Feature-based and visual explanation techniques like LIME, SHAP, Grad-CAM, DeepLIFT, and LRP are analyzed extensively. The authors note thatmany existing techniques for explaining AI decisions arepost-hoc in nature, and therefore, lack both reliability and coherence.Moreover,theprocessofinterpretingEEGdata is further complicated by domain-specific issues like high subject variability and low signal-to-noise ratio. Among other shortcomings, the study highlights problems related to inconsistent evaluation criteria and weak neuroscientific foundation of explanations provided by XAI models. Despite increasing their transparency, most of these techniques fail to achieve a reasonable level of alignment with physiology knowledge. In ordertoaddressthischallenge,thereportsuggestsfocusingon the development of more reliable XAI methods and validation of their results by experts [9].

Upadhya et al. [10] highlight the difficulties in detecting stressinworkplacesettingsduetotheshortcomingsassociated withconventionalself-reportingprocedures,whichareoften biased and unreliable. It is mentioned that physiological signs and behaviors serve as more accurate indicators of stress as they cannot be manipulated easily. The researchers point out the need for automated systems capable of providing timely intervention concerning psychological issues. The suggested approach offers a multimodal solution by incorporating physiologicalfeaturesalongwithfacedatainordertoimprovestress detection. Machine learning algorithms such as CNN models (VGG16 and Custom CNN), SVM, and LSTM are used to recognize both temporal and visual characteristics. According to the findings, LSTM proves to be more effective when analyzing physiological information while VGG16 delivers better results when detecting emotions from faces. Combining different sources of information improves the reliability and accuracyoftheproposedsystem.Inaddition,thestudyapplies explainableartificialintelligenceapproaches,includingSHAP, LIME,andPermutationImportance.Thesestrategieshelpgain insights into the reasoning of the model and detect potential biasesinpredictions.Theresearchsuggestsaconsiderableimprovementinthequalityofprediction,withVGG16achieving around 87% accuracy. Thus, the application of explainability techniques alongside multimodal learning appears to be an effective way of improving stress detection.

Based on Jaber et al. [11], the problem this research aimsto tackle is that of explaining AI algorithms utilized for stress predictionmodels.Whilethereexistmanymethodsemploying physiological factors for detecting stress in a subject, most modelsaretreatedlikea'blackbox,'makingitdifficult for professionals to trust their predictions. As noted by the authors, it is important for any AI model used for medical predictions to provide an understandable result. Therefore, the goal of the current paper is to create an explainable approach for generating predictions which will allow understanding the process. In order to make an algorithm more explainable, the authors suggest using an AI structure similar to those used in medicine – blood test reports. In such case, inputs for training machinelearningmodelswillbedatareceivedfromwearables such as body temperature, respiration, perspiration, heart rate, and muscle tension. By using Shapley Additive Explanations (SHAP),thecontributionofeachinputfactorcanbeanalyzed. This will provide researchers with valuable insights into the model's functionality and will allow developing a report. This report should contain the likelihood of stress, physiological parameters,referenceintervals,andimportanceofeachfactor. Thus, it will help the user understand what leads to stress according to the model. Balanced random forest classifier was chosen for the task. Cross-validation yielded an average F1 score of 0.78. The validity of the proposed approach was verified through both objective indicators and the opinions of specialists in the field.

AccordingtoEnkhbayaretal.[12],depressioncanbecalled oneofthekeyissuesintermsofmentalhealthacrosstheglobe since it impacts millions of people and leads to a number of undesirableoutcomes,includingmortalityandreducedquality of life. Notably, various questionnaires can serve as a helpful toolforassessingaperson'smentalconditioninpractice,but traditional approaches to depression diagnosis tend to have flawsassociatedwithsubjectivityandunreliability.Thus,there is a need for objective and automatic tools for depression diagnosis and identification.

Inparticular,thisresearchusesPSGphenotypesthatinclude physiological markers, clinical observations, and the individual's assessment of his/her mental state via questionnaires.The features used for detecting depression in this case include blood pressure, sleep efficiency, apnea–hypopnea index, anda mental well-being score measured through questionnaires. Four machine learning models were created on the basis of Random Forest, XGBoost, LightGBM, and CatBoost algorithms. An explainable artificial intelligence approach was utilized with the help of SHAP as well.

Theresultsshowthattheensembleapproachperformsbetter than others in this case since it provides up to 85% accuracy and the highest possible F1-score of 85%. Taking all these findingsintoaccount,itispossibletoclaimthattheutilization ofexplainableAItogetherwithPSGallowspredictingvarious mental conditions, including depression, and choosing the most appropriate treatment. [12]

III. PROPOSEDSYSTEM

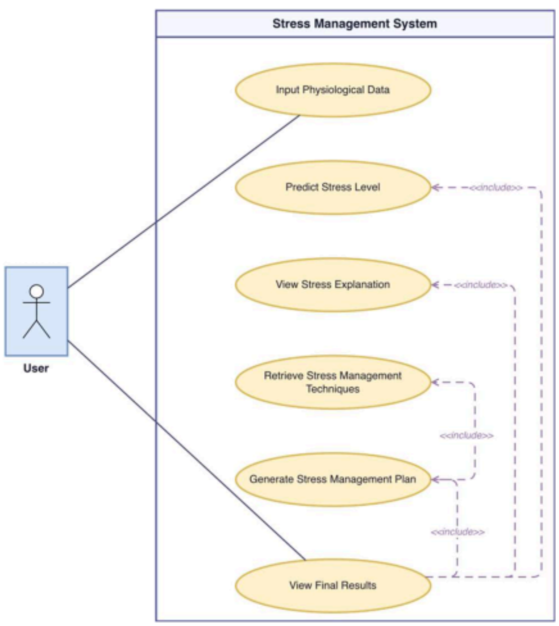

Fig.1.UseCaseDiagramoftheProposedSystem

Thedesignedmodeladoptsphysiologicalsignalstoachieve a holistic methodology in identifying stress and personalizing stress coping techniques. Using the physiological information gathered through EEG and ECG, the model is capable of evaluatingstresslevelsanddevisinginterventionplans.Figure 1 shows how the users interact with the system. The user provides input data regarding their physiological information, whichisusedbythesystemtoidentifysignificantfeatures with respect to the process of stress evaluation. The system thendetectsthekeycontributingfactorsandpredictstheuser's stress level based on the information provided by the users. This information is used to obtain stress relief techniques and develop personal plans.

IV. METHODOLOGY

A. ProblemFormulation

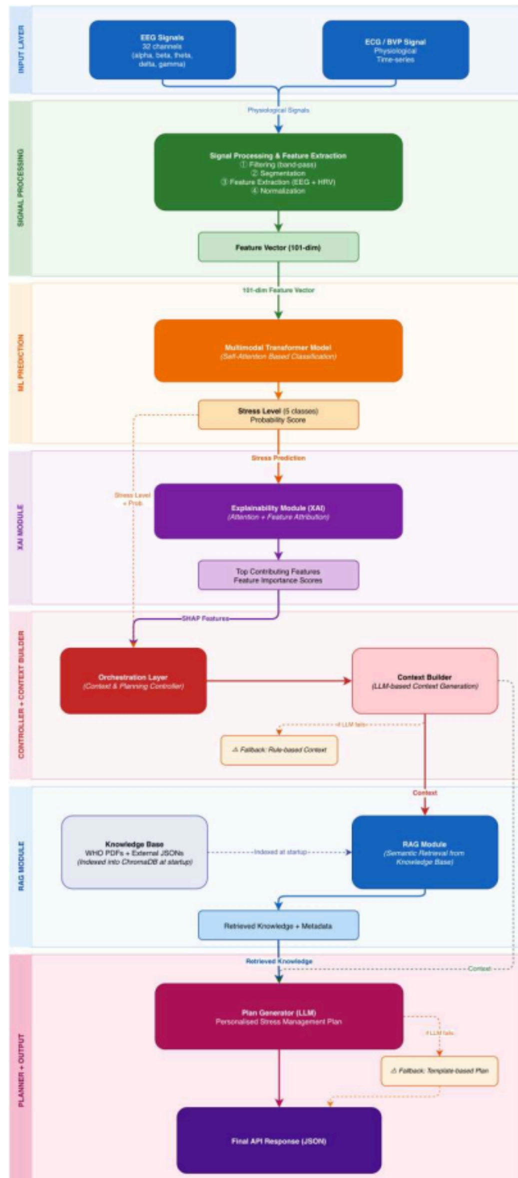

Fig.2.OverallSystemArchitectureoftheProposedSystem

The suggested model is a multimodal model that uses physiological signals for predicting stress level and providing customized stress management. The suggested model appliesa retrieval-based LLM model for generating the interventions plan, and it utilizes EEG and ECG features for categorizingthestressinfivelevelsofintensity.Figure2showstheo verall structure of the suggested model.

B. FeatureRepresentationandSignalProcessing

In order to create a single feature representation, the system analyzes physiological information from EEG and ECG sources.

- EEGSignals:Multi-channelrecordingsthatcapturebrain activity in theta, beta, and alpha frequency bands.
- ECG Signals: Heart rate variability (HRV) measures are derived from time-series physiological signals.

Feature extraction comes after preprocessing, which includes filtering,segmentation,andnormalizationoftherawdata. Asaresult,a101-dimensionalfeaturevectorisproduced that includes: 97 characteristics based on EEG and Meanheart rate, SDNN, RMSSD, and LF/HF ratio are the four ECG/HRV characteristics. The classification model receives this structured representation as input.

C. MultimodalStressClassificationModel

The stress levels are categorized through a multimodal Transformer architecture. The model takes into account the interrelationship between EEG and ECG data types through theuseofembeddingsandself-attentiontoprocessthefeature vector input.

The distribution of probabilities for five different stress classes is generated using the ability of the model to learn global representations and then pass them through the classification layer: very_low, low, moderate, high, very_high.

The stress level that has the maximum probability will be considered.

The highest likelihood score is used to establish the final stress level.

D. ExplainabilityModule

AnXAImoduleisimplementedwithinthearchitecture inordertoenhancetheinterpretabilityoftheresults.The XAI module explains the model's predictions by leveraging two different approaches: the first being an attention-based approachthatemphasizestheimportanceoftheinputfeatures; the second being a permutation-based approach that measures the importance of each feature.

E. ContextConstructionandOrchestration

Thepredictionresultsgiverisetoanenvironmentcreated via an orchestration layer. Such an environment includes

- stressprobabilityandassociatedlevel
- important physiological factors contributing to

stressThisinformationisrenderedtoareadabledescriptionof stressbythecontextbuilderandbecomestheinputtosubsequentstages.

Inordertokeepthesystemrobustincaseoflanguagemodel unavailability, arule-based fallback approachis used tocreate a reduced environment.

F. Retrieval-AugmentedKnowledgeIntegration

RAG framework is used for the integration of external knowledge.Avectordatabasehasavettedknowledgebase of stress-reduction strategies.

The system uses semantic similarity-based retrieval to find pertinentinterventionoptionsgiventhecreatedcontext.These includeresearch-provenmethodsthatcomplementwell-known strategieslikemindfulness-basedstressreduction(MBSR)and cognitive behavioral therapy (CBT).

G. PersonalizedPlanGeneration

Acustomizedstressmanagementstrategyiscreatedusing a large language model (LLM). The input for the model is:

- Theconstructedstresscontext
- Retrievedinterventiontechniques

Thesystemusesthisdatatocreateorganized,practicalsug- gestions that are specific to the user's stress level.

Atemplate-basedfallbackmethodusesretrievedtechniques to create a viable plan in the event that the language model fails or is unavailable.

H. OverallWorkflow

A sequential pipeline is used in the suggested system. A multimodalTransformermodelusestheinformationextracted from physiological inputs to forecast stress levels. Explainabilitytechniquesareusedtointerpretthepredictions,and a structured context is produced. A large language model (LLM) generates a customized stress management strategy, and a knowledge-based retrieval system retrieves pertinent intervention options. The result is a structured response with suggested actions and stress forecasts.

V. Results

The proposed multimodal stress management system successfully carried out the end-to-end pipeline, transforming the inputof101-dimensionalphysiologicalfeaturevectorsinto a five class stress prediction (very low to very high). The multimodal transformer produced probabilities for the five stressclasses,wherethemaximumprobabilitywasconsidered as the stress confidence score. In addition, the system pro- vided solid explainability signals in the form of normalized feature importance scores for each of the 101 features and cross-modal attention weights between CLS-token. The latter outputs enabled the necessary physiological explanation and reasoning for subsequent intervention planning phases, giving JSON outputs suitable with API contracts.

Inordertogroundthepersonalizedinterventionrecommendations on proof based approach, the proposed framework made use of the knowledge base built in ChromaDB to retrieverelevantinterventiontechniquesbasedontheirsemantic similarities. The RAG pipeline consistently produced well structuredcontextualmetadataconsistingoftitles,duration, procedure, and applicability for different stress levels. Based on this retrieved evidence, the main Groq language model plannerconstructedpersonalizedinterventionactionsets,combiningstress-awarecausationwithphysiologicalinterpretation. From a qualitative standpoint, the resulting action sets werein line with predicted stress levels: very high-stress contexts suggested fast autonomic regulation techniques (grounding, breathing), medium – cognitive-behavioral routine management, and lower – resilience building.

The proposed multimodal system showed a high level of resiliency and maintainability in the face of simulated limitations. As a part of mitigating the risks of an unavailable external language model, a fallback plan was designed. Based on theretrievedcontextualinformationfromChromaDBevidence base, the deterministic fallback planner produced valid and schemacompliantinterventionactionplans,whichallowedthe pipeline up and running. Although such a pipeline has much simpler reasoning compared to the one based on LLMs, it is moreresilienttofailuresandproducesresultscompatiblewith API contracts.

VI. Discussion

A major novelty of the study presented above relates to translating a multimodal stress pipeline into an actionable clinical context through decision support. As opposed to a majorityofstudiesfocusingonclassificationperformance,we introduce an approach that bridges prediction with guidanceon actionable steps to manage stress levels in an individual patient.Bycombiningpredictionsofstresslevelsusingcentral nervoussystemmetrics(EEG)andautonomicstressindicators (ECG/HRV) with an interpretability component, our model provides physiological justification for stress categories and potential reasons for these predictions. In addition, the combinationof101featuresinonemodalityallowssynthesizing an explanation-driven rationale that will result in a set of recommendations based on the patient's physiology insteadof general clinical practice guidelines.

One of the advantages of our solution comes down to leveragingthepowerofRAGtolimitinterventionselectionto only those that correspond to a clinically informed database. The use of RAG helps mitigate safety concerns related toLLM uncurated outputs. Furthermore, our approach carefully accounts for the aforementioned trade-off of personalization vs. high availability by offering a deterministic fallback strategy for cases when the Groq LLM is unavailable. The latter guarantees graceful degradation in case the original LLMfails. To provide safe human oversight, the system also offers transparency about the source of each recommendation by providing a corresponding plan_source flag in case the output isgeneratedviathefallbackstrategy.Suchstratificationwould help differentiate between LLM-specific reasoning and robust high-availability service in terms of the final intervention quality.

VII. Conclusion

In this paper, we propose a complete multimodal pipelinetoaddresspersonalizedstressdetectionandmanagement. We integrate EEG metrics along with ECG data into 101 dimensional features vector, which can be used as input to a transformer-based model to classify individuals' stress level with great precision in terms of five-levels categorization.The most important contribution of our solution is going beyond the predictive classifier and moving forward to a clin- ical decision support model with clinical reasoning abilities. An explainable AI module will provide interpretability by highlighting the physiological markers through cross-modal attention. Furthermore, we ensure that the generated solutions are grounded in evidence-based therapeutic approaches, such as cognitive behavioral therapy or mindfulness based stress reduction, through a RAG process. Finally, a deterministic fallback algorithm ensures uninterrupted operation in case of unavailability of large language models APIs.

VIII. FutureWork

In order to advance this framework, future research should look towards clinical validation through trials conducted in clinical environments and work environments where the solutionistested.Whilethisoccurs,theabilitytoprocessmultiple modes of inputs should also be increased by incorporating more sensor based data from alternative modes. Including reinforcementlearningthroughafeedbackmechanismwillallow for continuous improvement in personalizing the generated plans.

Stress Detection using ECG-EEG

ORIGINALITY REPORT

1%	1%	0%	0%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	everydaypsy.com Internet Source	<1%
2	Submitted to uvt Student Paper	<1%
3	www.indianpac.com Internet Source	<1%

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	Off		